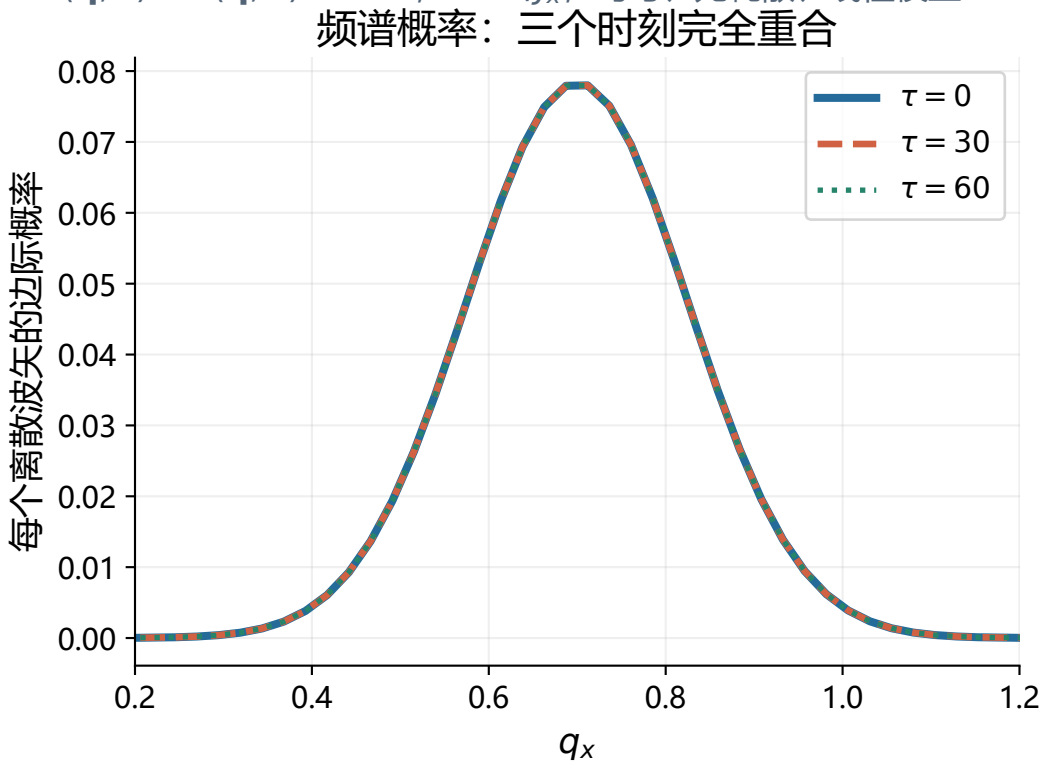
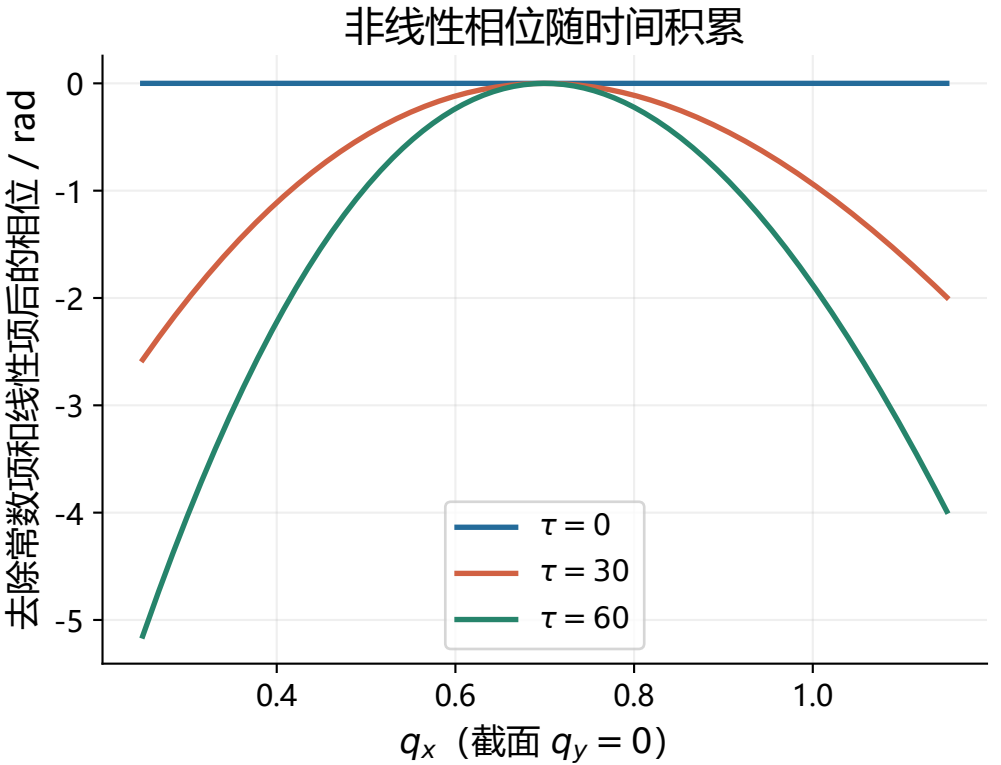
C12 波包变形来自相位积累，频谱概率保持不变

$A(\mathbf{q},\tau)=A(\mathbf{q},0)e^{-ie(\mathbf{q})\tau}$，$e=\varepsilon/J_x$；均匀、无耗散、线性模型





右图去除了整体相位与中心群速度产生的线性相位，剩余相位反映色散引起的形变。